

Adrian Legaspi

Backend Engineer

Monterrey, Mexico

Profile

Senior Software Engineer with 9+ years delivering API-first backend services, event-driven serverless integrations, and semantic data platforms in Node.js, TypeScript, and Python, across enterprise security, fintech, and agentic AI products.

Designs ontologies and semantic data models for enterprise security data, formalized as OWL class and property hierarchies covering domain taxonomies and relationship modeling. Builds LLM applications end to end: RAG retrieval over Chroma vector search, Model Context Protocol integration, agent orchestration, prompt engineering, and fine-tuning workflows, deployed on AWS with Docker and CI/CD. Professional-level English for technical design discussions, documentation, and cross-functional collaboration.

Experience

Software Engineer, JointechAI – Tijuana, Mexico

Aug 2022 – present

- **Viasat Internal Security Application (via Borderux)**: Designed and maintained the platform ontology and semantic data model, formalized as an OWL class and property hierarchy: products, the findings raised against them, and the assets that discover those findings. Government-contract security program; specifics under NDA.
- Maintained and evolved that taxonomy as the domain grew, keeping products, findings and the assets that discover them queryable as one connected model instead of per-source silos.
- Integrated heterogeneous scanner and asset sources into that model through event-driven Lambda handlers, S3 event triggers, and inbound webhooks, with Redis caching on hot read paths and CloudWatch logs and alarms for operational visibility.
- Delivered Python AWS Lambda functions and REST API services behind API Gateway, each deployed and versioned independently, embedded in a backend-heavy team of 15.
- Built the centralized React and TypeScript dashboard that aggregates security findings for 4,000+ internal users, as the sole frontend engineer on the team.
- Worked across the platform AWS environment, including S3, EC2, CloudWatch and Lightsail, with releases shipped through CodePipeline and GitHub Actions.
- **Agentic HR ERP (Jointech AI)**: Tech Lead for a team of 5 engineers on the internal HR platform managing 30 employees and analyzing prospective roles and candidates for hiring.
- Built its Node.js, Express and TypeScript backend: REST API design, service integration, and the data model behind it.
- Implemented the RAG architecture covering document ingestion, chunking, embeddings and Chroma vector search, delivering grounded semantic search across heterogeneous HR and candidate documents.
- Developed agent orchestration and prompt-engineering workflows over Model Context Protocol, wiring retrieval and structured tools into LLM calls, plus model fine-tuning workflows. Built the React client on top of those services.

Software Engineer, Arte Digital – Tijuana, Mexico

Jan 2020 – Aug 2022

- **Captira**: Designed and built secure Laravel REST APIs serving 3,000+ legal professionals, with the relational data model behind them.
- Built complex Angular data dashboards on those APIs, covering case, document and reporting workflows.
- **Crediverso**: Architected and shipped full-stack fintech features for a platform serving 10,000 users, on React and a headless WordPress backend.
- Established the CI/CD pipelines with Codeship, cutting deployment time for the team.

Web Designer & Developer, LuballSoftware (Freelance) – Tijuana, Mexico

Mar 2016 – Jan 2020

- Developed and managed FireApp, a digital logging application for the Tijuana Fire Department used by 180+ field firefighters and paramedics across 18 stations for pre-hospital incident reports.
- Delivered 25+ web solutions for local businesses over 4 years, ranging from custom e-commerce platforms to landing pages.

Education

Universidad Tecnológica de Tijuana, BS in Tecnologías de la Información – Tijuana, Mexico

May 2015

Skills

Ontologies & Semantic Data: OWL ontologies, semantic data models, taxonomies, class and property hierarchies, domain modeling

Backend & API Engineering: Node.js, Express, TypeScript, Python, REST API design, API-first contracts, AWS Lambda, API Gateway, PostgreSQL, Redis, Firebase, Laravel (PHP)

Architecture & Integration: Serverless microservices, event-driven handlers (S3 events, webhooks), asynchronous workflows, service-to-service integration, data integration patterns, Model Context Protocol

Generative AI & Semantic Search: RAG architectures, Chroma vector search, embeddings, semantic retrieval, agent orchestration, prompt engineering, fine-tuning, Gemini APIs

Cloud & DevOps: AWS (Lambda, API Gateway, S3, EC2, CloudWatch, Lightsail, CodePipeline), Docker, CI/CD (GitHub Actions, CodePipeline), software architecture

Frontend: React, TypeScript, Angular, SCSS

Languages

English: Professional proficiency

Spanish: Native

Side projects

Shroom Merge ([Play Store](#), [App Store](#))

July 2026

- Independently developed and released Shroom Merge (Kinoko Merge on iOS), a mobile merge game, taking the project from concept to Play Store and App Store launch.